Module 14 Lecture - Replication and Transparency

Research Methods in Psychology

Quinton Quagliano, M.S., C.S.P

Department of Psychology

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1 Chapter Overview

1.1 Learning Objectives

- Explain why it is essential for a study to be replicated.
- Describe why transparent research practices help ensure credible science.
- Evaluate, in a nuanced way, a study's quality in the context of scientific progress and external validity
- Appreciate science as an ongoing and continuous process of improvement, guided by replication

1.2 Chapter Overview

- As we've learned over the course of this class, _____ of scientific work can look/work a lot of different ways.
  - This is often different based on the _____ of the study as well - experiments v. small-n v. bivariate, etc.
- In short, valid studies are those that provide enough _____ that a finding represents a real effect, change, or description - not some fluke.
  - Said fluke could come from a failure in any of our 4 _____
- One of the best ways to _____ our evidence is to demonstrate that our findings can be produced again under the same or similar conditions.

2 Replication

2.1 Overview

? In which validity have we previously described reproducability?

- (●) A) Statistical
- ( ) B) Internal
- ( ) C) External
- ( ) D) Construct

Explanation:

- Replication/reproducability is where we determine whether a finding can be reproduced or if it appears to be a _____
  - It can be viewed as a general indicator of a _____ quality study
- *Recall:* Our findings are often judged against $\alpha = 0.05$, which means we do leave some room for "random _____."
  - By that logic, our results could simply be _____ due to chance
- Often, we evaluate _____ as being of core importance to a study's value - because we want to know if the results tell a consistent story
  - Remember that we look for _____ in the evidence body, not just one-off studies

2.2 Types of Replication

- "Replication" is a fairly loose term, and actually applies to many scientific _____.
  - You have likely run across _____ studies while you looked for articles on your research proposal.
- In a replication study, our _____ is not to refute or support, our motivation is to honestly explore and assess the phenomenon
  - "Trust but verify"
- We may separate study replications into a few different types:

2.2.1 Direct

- This is the _____ type of replication, where we attempt to largely keep all factors, measures, and circumstances consistent with the original study.
  - This is effectively following the _____ described in a journal article *exactly* as it is written.

Discuss: What issues could make a direct replication study harder to pull off?

- It is, of course, done on a new _____, but from the same theoretical population.
- Design issues are _____ forward in direct replications, as we attempt to stay largely the same (even when that may be a bad thing)
- *Example:* Imagine if I re-did the Harlow Monkey study *exactly* the same way Harlow did it.

2.2.2 Conceptual

- Our central hypothesis stays the same as the _____ study, but we may change minor procedures, designs, and how we operationalize our latent constructs.
  - But, in a conceptual replication, we will not _____ anything new.
- This can be a useful way to _____ some glaring validity issues in an original study and do a more refined investigation
- However, it should be viewed critically as a _____ version of the original, so corrections to the original's effects may not be as clear
- *Example:* Harlow's Monkey's again, but this time I've changed the food with the wire mother to be more appealing and I have turned up the heating element on the felt mother just a little bit more

2.2.3 Replication and Extension

- In this type, we add _____ conditions or variables to add more nuance and context to findings.
- Such procedures could reveal _____ and mediators, or just present other relevant outcomes for convergence

- Like any replication, this type may offer evidence to _____ the findings of the original study, and also add their own new information at the same time

📢 Discuss: Do you think your research proposal is one of the 3 types of replications? If so, which one?

- *Examples:* Harlow's Monkeys, but instead, I am also going to add baby gorillas in to see if their behavior differs.

2.3 Real Projects

- A replication effort/project can look different ways, but is almost always very time-consuming and _____.
- It is beneficial to have multiple _____ and sets of researchers replicate an effect to reduce the chance that a single experimenter's bias affects results.

2.3.1 One Study, Many Labs

- In this scenario, one research lab "leads" numerous other labs in applying replications to a single central _____ and research design.
- Then, all labs' _____ are looked together at once and examined for their convergence or divergence.
- Because the scope of such an effort tends to be very _____ and focused on just one original study, we are more likely to see direct and, to a lesser extent, conceptual replications.

2.3.2 Many Studies, Many Labs

- This is largely an _____ of the above type, where we now focus on a broader issue or topic and try different studies within that area.

- A project such as this adds value by approaching a hypothesis with a variety of conceptual and extension replications, to see how different outcomes and _____ may contribute to significant findings (or not)

📢 Discuss: Why not just do this type with every important study? What challenges do we run into?

2.3.3 When It Doesn't Work

- Some replications find _____ or even opposite results from the original study!
- One possibility is that the two studies differ in some notable, _____ way, i.e., if you change many things, it is no surprise that the results are different
- Or, either the _____ or original study is flawed or a fluke
- So what next? Replicate some more! - A single-_____ replication is likely to not be sufficient to full dissuade the scientific audience from the original finding.
  - This is the advantage of multi-_____ studies - it helps ensure that reproduction is consistent or trends in a certain direction

2.4 Meta-analysis

- Instead of embarking on running many replications ourselves, we may choose to _____ summarize the existing evidence.

? Meta-analyses are technically a subtype of what paper type?

- (●) A) Empirical original journal article
- ( ) B) Scientific journalism
- ( ) C) Literature review
- ( ) D) Replication

Explanation:

- b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B204D657461203E7D7D–analyses quantitatively averages results across many similar studies to determine an aggregate/composite average.
- The averaging can be done with any _____ of effect size, such as Cohen's d and r.

2.4.1 Strengths and Limitations

- Meta-analyses can be an _____ way to give a summary of certain area with a clear number, informing future researchers of the state of the science at that time.

Discuss: Meta-analyses tend to be very popular to cite, why do you think this is?

- However, this _____ lives and dies by its attention to detail and finding all relevant studies.
- Especially, null findings may be subject to the "_____ drawer problem", and may be obscured from being collected and aggregated.
  - Most meta-analyses do recognize the role that publication bias can play in results

2.5 In Popular Culture

- Rarely, does popular journalism adequately capture the _____ and changing nature of science.
- Journalism is also partially drawn to _____ and new research, but may give less emphasis to historical, but still relevant, studies.

  - Replications tend to be somewhat less interesting, but still have very important results.
  - "If it bleeds, it leads"

- In general, it is much better to start with a peer-reviewed, scientifically _____ meta-analysis or literature review when orienting to a new area.

3 Transparency and Credibility

3.1 Overview

- Even well-meaning scientists (but also malicious actors) can make _____ in truthfully reporting results

> Discuss: Why do you think researchers publish research that is faked or misleading?

- It should always be our goal to make b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B207468656F7279203E7D7D–driven predictions and designs - not making changes just for beneficial results.
- We have a variety of questionable practices to _____ and transparent practices to try to stick to which aid in reproducability

3.2 Questionable Practices

3.2.1 Under-report Null Findings

- "Real" research tends to involve more than one outcome variable of interest, and sometimes more than one _____
- But, some authors may downplay many null _____ in favor of focusing on the shiny significant findings.

- This creates a narrative of _____ importance and reproducability of findings, when the majority of evidence in a study actually points to the contrary.

3.2.2 HARKing

- This stands for "Hypothesis After Results are _____"
- Generally, we want to make predictions and hypotheses prior to our data _____ and analyses
- As a general ethical rule, it is bad to say "I knew it all along!" - as it is not _____ or literature driven.

3.2.3 P-hacking

- This occurs when multiple analysis methods are used _____ a significant p-value is found, and usually only the final method is reported
- Similar to under-reporting null findings, this tends to _____ contrary evidence and overstate otherwise weak findings
- Early decisions on design, hypotheses, and analyses need to be guided first-and-foremost by _____

? Review - what is it called when someone changes data to fit their hypothesis better?

- (●) A) Fabrication
- ( ) B) Falsification
- ( ) C) Fixing
- ( ) D) Flubbing

Explanation:

3.3 Transparent Practices

- To _____ the previous, questionable techniques - we may use

a few strategies to increase public and scientific confidence in our results.

- Often, these "open" methods intentionally make us _____ to wide critique from the broader public - but this is good for science!

3.3.1 Open Data and Open Materials

- Open _____ is when we publicly share anonymized data that can be re-analyzed by anyone to ensure results were correct
- Open materials provide all in-depth procedures, measures, and tools to be used in any _____.
- Both of these may be posted on repositories like https://github.com or https://osf.io

> Discuss: What circumstances might prevent releasing data and materials to the public?

3.3.2 Preregistration

- As previously mentioned, this is the _____ announcement of one's hypothesis and plans prior to completing a study.
- Doing this helps _____ any desire to HARK or p-hack, as any interested party could easily point out that you changed your future analysis.

4 When Do We Need (Really) External Validity

4.1 Overview

- Replication (especially _____) can help us establish just how widespread and generalizable effects are outside just our original sample of individuals.

- As we _____ original studies, we are able to better see just what people and contexts they can apply to
- External validity is one of the most _____ and difficult to establish validities - therefore we must discern how to balance our priorities with it

4.2 Generalization to Other People

- Remember that our sample comes from a _____ population that we chose, not just everyone in general
- Samples are primarily strong by virtue of their _____, not only their size!
- Samples can be easily _____ by convenience - think carefully about the impact that it is likely to have!

? Review - which of the following techniques results in an unbiased sample?

- (●) A) Quota sampling
- ( ) B) Convenience sampling
- ( ) C) Self sampling
- ( ) D) Cluster sampling

Explanation:

- Results may not generalize to _____, but we must consider whether this is vitally important to us.

4.3 Generalization to Other Settings

- Many b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B206C6162203E7D7D–based studies may benefit from replication in more “real-world” settings
- On the other hand, we may be interested to see if we can isolate a _____ behavior within the confines of the lab
- More broadly, we likely also want to examine how our studies look outside of the geographic and _____ areas they are conducted.

4.4 Do We Always Need to Generalize?

4.4.1 Theory-testing Mode

- This is the "first" stage, where we attempt to _____ and describe a theory accurately under more controlled conditions
- We work knowing external validity is _____, but taking care to establish other strong validities.
- In this scenario, often _____ validity is treated as most important (if we have an experiment)
- _____ claims usually work in a "theory-testing" mode.

? Early in the semester, we gave types of research based on their scope, similar to the present discussion - which of the following was not one of those types?

- (●) A) Basic
- ( ) B) Supplementary
- ( ) C) Applied
- ( ) D) Translational

Explanation:

4.4.2 Generalization Mode

- _____ and association claims more often aim to be widely applicable and relevant.
- Generalization mode places a special emphasis on making sure theories have strong _____ validity, putting less focus on internal, like when theory-testing
- _____ psychology is a sub-discipline almost entirely determined to investigate cross-cultural differences and similarities, working in generalization mode.

4.5 Do We Always Need to be in the Real World?

- Valuable research happens in _____ setting, both lab-controlled, and completely ecological (i.e., the “real world”)!
- Researchers in these separate settings often _____ greatly from the mutual shared knowledge.

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms